Sumanth Mamidi

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SUMMARY

Second-year B.Tech CSE (AI & ML) student with hands-on experience building applied ML and LLM agent systems: vision-transformer ocean modeling, YOLOv8 vehicle tracking, citation-verified RAG pipelines, and multi-agent security repair. Author of a published research paper. Seeking an AI systems, ML or full-stack internship.

EDUCATION

SRM Institute of Science and Technology, Ramapuram, Chennai — Aug 2025 – May 2029
B.Tech, Computer Science and Engineering (AI & ML) — **CGPA: 10.0/10.0**

Sri Sarada Junior College, Vijayawada — Higher Secondary Certificate (MPC), **98.2%** — Jun 2023 – Mar 2025

S.T.B.E.M High School, Vijayawada — Secondary School Certificate (SSC), **93%** — Jun 2022 – Apr 2023

PUBLICATIONS

L. Y. Reddy, **M. N. S. Sumanth**, R. Regin, S. R. Bose, S. S. Jeev, and S. B. Sherine, "SECUREPATCH: Specialized Multi-Agent Architecture with Automated Validation for Security Vulnerability Repair," *AVE Trends in Intelligent Computing Systems*, vol. 3, no. 1, pp. 23–47, 2026. DOI: 10.64091/ATICS.2026.000283.
Contribution: initial multi-agent prototype (OMNIFIX). The paper reports 99.4% repair success on 160 vulnerabilities.

PROJECTS

OceanEmbed — 3D Ocean Temperature Reconstruction — github.com/SumanthMamidi-MNS/OceanEmbed
PyTorch · Vision Transformer · FastAPI · React

- Predicts 3D ocean temperature (15 depths, to 1,000 m) from satellite data with a CNN–Vision Transformer.
- 0.98 °C thermocline error, 33% below climatology, validated on 5,512 Argo float profiles.

UrbanTrace — Multi-Camera ANPR Vehicle Tracking — github.com/SumanthMamidi-MNS/UrbanTrace
YOLOv8 · Min-Cost Flow · FastAPI · WebSockets · Docker

- Tracks vehicles across city cameras by fusing noisy plate reads, appearance, and travel time (0.972 IDF1).
- Raised degraded-plate accuracy from 87.8% to 99.95%; flags cloned plates in real time (0.998 AUC).

TrueCite — Citation-Verified Legal RAG — github.com/SumanthMamidi-MNS/TrueCite
FastAPI · ChromaDB · BGE-M3 · BM25 · Knowledge Graph

- Legal Q&A that cites its sources: hybrid BGE-M3 + BM25 retrieval over 20 legal instruments.
- Checks every claim with 2-of-3 entailment consensus: zero false answers on benchmarks.

FieldCast — Hyperlocal Agro-Weather Downscaling — github.com/SumanthMamidi-MNS/FieldCast
LightGBM · FastAPI · React · MapLibre GL · TypeScript

- Downscales block forecasts to 9,802 gram panchayats with a two-stage LightGBM quantile model.
- Calibrated 80% prediction intervals checked against NOAA gauges; NumPy-only inference in ~100 ms.

Auxin — AI-Powered Agricultural Commerce Platform — github.com/SumanthMamidi-MNS/Auxin
React / Next.js · FastAPI · PostgreSQL / Supabase · RAG

- Full-stack farm-to-customer platform with customer, farmer, delivery, and admin portals.
- GPS job matching with OTP-verified delivery; Milo AI answers agricultural questions via RAG.

TECHNICAL SKILLS

Languages	Python, JavaScript, TypeScript, SQL
ML & Deep Learning	PyTorch, scikit-learn, LightGBM, Vision Transformers, YOLOv8, OpenCV, NumPy
LLMs & RAG	Claude and Gemini APIs, Ollama (Llama 3.1, Qwen 2.5), RAG, ChromaDB, BGE-M3
Backend & Web	FastAPI, REST APIs, WebSockets, React, Next.js, Tailwind CSS, MapLibre GL
Data & Tools	PostgreSQL, Supabase, NetCDF / Xarray, Docker, Git, GitHub

CERTIFICATIONS & HACKATHONS

Google (Coursera)	AI Professional Certificate · AI Essentials
Anthropic	Claude API · Claude Code · Model Context Protocol · Agent Skills · Subagents
Forage & Scaler	BCG X GenAI · Walmart Software Engineering · Multi-Agent Systems · RAG · MLOps
Hackathons	Participant: Adobe University Hackathon · HackerRank 24-hour hackathon